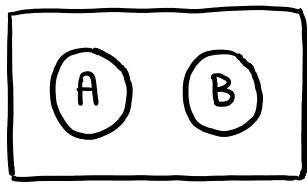


MUTUALLY EXCLUSIVE EVENTS:

$P(A \cap B) = P(\emptyset) = 0$
 $P(A|B) = 0$
 $P(B|A) = 0$



mutually exclusive events are not independent.

$P(A|B^c) = \frac{P(A \cap B^c)}{P(B^c)} = \frac{P(A)}{P(B^c)}$

If $A \subseteq B$, then they are neither mutually exclusive nor independent.

Zero probability events are always independent.

$P(E) = 0, P(F) = 0$
 $E \cap F \subseteq E \Rightarrow P(E \cap F) \leq P(E)$
 $P(E \cap F) = 0$

CONDITIONAL INDEPENDENCE:

$P(A|BC) = \frac{P(AB|C)}{P(B|C)} \Rightarrow P(AB|C) = P(A|BC) \cdot P(B|C)$

Two events A and B are conditionally independent of C if:
 $P(AB|C) = P(A|C) \cdot P(B|C)$

As a consequence, $P(A|BC) = P(A|C)$

This does NOT imply A and B are independent.

PRINCIPLE OF COUNTING

Given n objects, in how many ways can you arrange them?
 $n!$

${}^n P_r = \frac{n!}{(n-r)!} \quad \Bigg| \quad {}^n C_r = \frac{n!}{(n-r)! r!}$

Given n objects, how many different pairs can you form?
 ${}^n C_2$

- Sampling a set = choosing an element from the set
- Sampling uniformly at random
= all items have equal probability of being chosen.

sampling $\begin{cases} \rightarrow \text{with replacement} \\ \rightarrow \text{without replacement} \end{cases}$

sampling $\begin{cases} \rightarrow \text{ordered} \\ \rightarrow \text{unordered} \end{cases}$

CHOOSING K OUT OF n OBJECTS

- Ordered sampling with replacement: n^K
- Ordered sampling without replacement: ${}^n P_k$
- Unordered sampling without replacement: ${}^n C_k$
- Unordered sampling with replacement:

Each object i can be chosen at most k times

$0 \leq x_i \leq k$
 $\sum_{i=1}^n x_i = k$

${}^{n+k-1} C_k$

We have k balls.
(n-1) separators
no. of possibilities
 $= {}^{n-1+k} C_{n-1} = {}^{n-1+k} C_k = {}^{n+k-1} C_k$

balls $\rightarrow$ separators $\rightarrow$
• string of length (stars+bars).
• no. of permutations
 $\Rightarrow \frac{(\text{stars+bars})}{(\text{stars})! (\text{bars})!} = {}^{\text{stars+bars}} C_{\text{bars}}$

Q: There are n red balls and r bins.
How many ways can you put these balls in bin such that no bin is empty?

ans: No. of solutions to $\sum_{i=1}^r x_i = n, x_i > 0$

Arrange n balls in a line.
We have (r-1) separators.
 $\therefore$ n-1 spaces between n balls to place r-1 separators.
 $\Rightarrow {}^{n-1} C_{r-1}$

$\sum_{i=1}^r y_i = n-r, y_i \geq 0$
(n-r) balls
(r-1) separators
no. of possibilities
 $= {}^{n-r+r-1} C_{r-1} = {}^{n-1} C_{r-1}$

What if bins can be empty?

= No. of solutions to

$\sum_{i=1}^r x_i = n, 0 \leq x_i \leq n$

Substitute $y_i = x_i + 1$

$\sum_{i=1}^r y_i = n+r, 0 < y_i < n$
 $\Rightarrow {}^{n+r-1} C_{r-1}$

n balls
(r-1) separators
no. of possibilities
 $= {}^{n+r-1} C_{r-1}$

Q: Toss a biased coin n times.
p=probability of head.
What is the prob. of k heads?

ans: ${}^n C_k \cdot p^k \cdot (1-p)^{n-k}$ $0^0 = 1$

Q: What is the probability that you will get head for the first time at the rth toss? $r \leq n$

ans: $(1-p)^{r-1} \cdot p$

Q: Roll a dice n times.
Prob. that half of them show 1 and the remaining half show 6?

ans: ${}^n C_{n/2} \times \left(\frac{1}{6}\right)^n$ ${}^n C_{n/2} \cdot \left(\frac{1}{6}\right)^{n/2} \cdot \left(\frac{1}{6}\right)^{n/2}$

What if n1 of them show 1 and n2 of them show 6?

$\frac{n!}{n_1! n_2! (n-n_1-n_2)!} \times \left(\frac{1}{6}\right)^{n_1} \left(\frac{1}{6}\right)^{n_2} \left(\frac{4}{6}\right)^{n-n_1-n_2}$